

DATA ANALYSIS INTERNSHIP

Passenger Manifest Analysis

Data Quality, Cleaning and Insights on 418 Records

Data Quality • Imputation • KPIs • Insights • Interactive Dashboard

Prepared by

KOUAME KOFFI FIDÈLE

September 19, 2026

Contents

1	Executive summary	2
2	Data quality	2
3	Cleaning and imputation	3
4	Indicators	4
5	Insights and recommendations	5
5.1	The target column is unusable, and that is the headline	5
5.2	Three quarters of the cabin column is missing, and the gap is the signal	5
5.3	Fare is the clearest description of the ship	5
5.4	Most passengers travelled alone	5
5.5	One age in five had to be reconstructed	6
6	The interactive dashboard	6
7	Limitations	9
	Deliverables	9

1 Executive summary

The brief asks for data cleaning, imputation, KPIs, insights and an interactive dashboard, using Python and nothing else. All five are delivered, and the tool constraint is respected at every stage.

The file arrives in good superficial condition: no duplicate rows, no duplicate identifiers, correct types throughout. It is still not fit for the analysis a reader would expect, and the reason is a defect no completeness audit detects.

Survived is an exact copy of Sex. All 152 women carry 1 and all 266 men carry 0, on **100.0% of rows**. Two of the four cells of the contingency table are empty.

This is *target leakage*: a column that should be predicted is derivable from another. A model trained on it scores perfectly and has learned nothing. Producing such a model would have been the easy path. This report does the opposite, and the consequence runs through every section: **no survival rate by any segment is reported**, because each would restate the sex split of that segment and nothing more.

2 Data quality

Table 1: Checks performed

Check	Result
Rows	418
Duplicate rows	0
Duplicate passenger identifiers	0
Missing Age	86 (20.6%)
Missing Fare	1
Missing Cabin	327 (78.2%)
Survived derivable from Sex	100.0%
Rows deleted	0

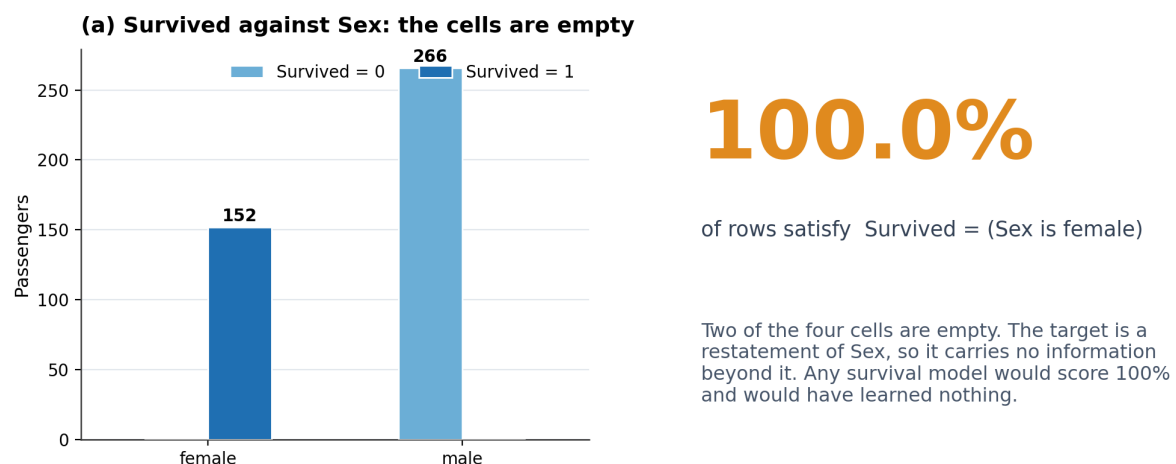


Figure 1: The target against sex. Half the canvas is given to a single number because that number invalidates an entire class of analysis a reader would otherwise expect.

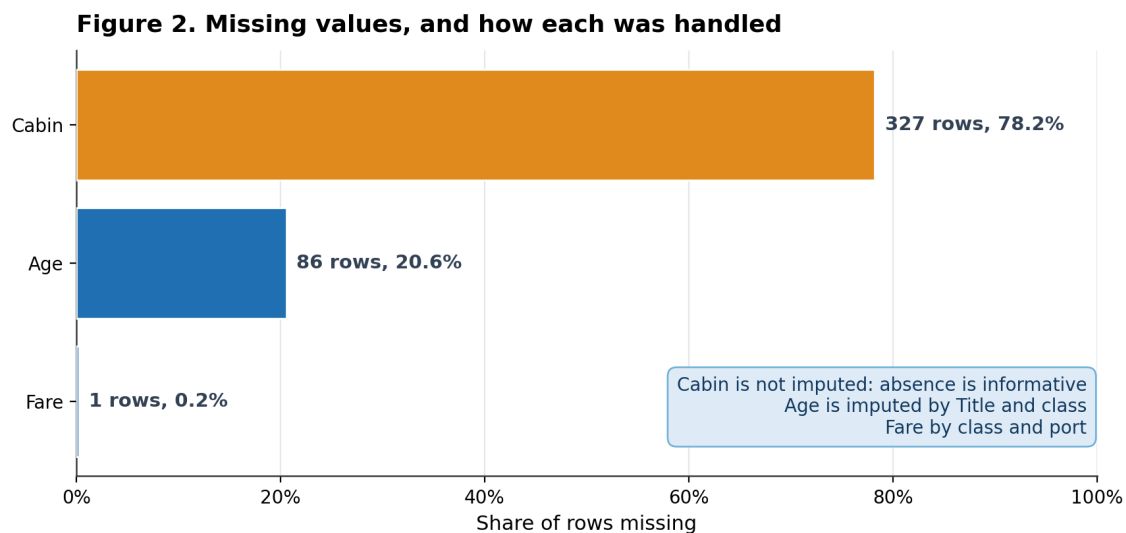


Figure 2: The three incomplete columns, and the decision taken on each.

3 Cleaning and imputation

Table 2: One decision per column, with its reason

Column	Missing	Decision and reason
Age	86	Imputed by the median of <code>Title</code> and <code>Pclass</code> together. <code>Title</code> carries life stage and class correlates with age, so the pair is far more specific than either alone. A global median would give a three-year-old the age of an officer
Fare	1	Imputed by the median of <code>Pclass</code> and <code>Embarked</code> , because fare is set by class and route
Cabin	327	Not imputed. Absence is informative: a number was recorded for 74.8% of first class against 1.8% of third. Inventing 327 values would fabricate 78.2% of a column
Zero fares	2 rows	Kept. One belongs to the chairman of the line, travelling on company business. A zero fare is a commercial arrangement, not a recording error

Every imputed value is flagged in `Age_imputed` and `Fare_imputed`, so a later reader can exclude them and see how far the conclusions move. **No row was deleted**; twelve columns become 24.

4 Indicators

Table 3: The manifest in seven figures

Passengers	418
Female share	36.4%
Median age	26.0 years
Median fare	14.45
Third class	52.2%
Travelled alone	60.5%
Cabin recorded	21.8%

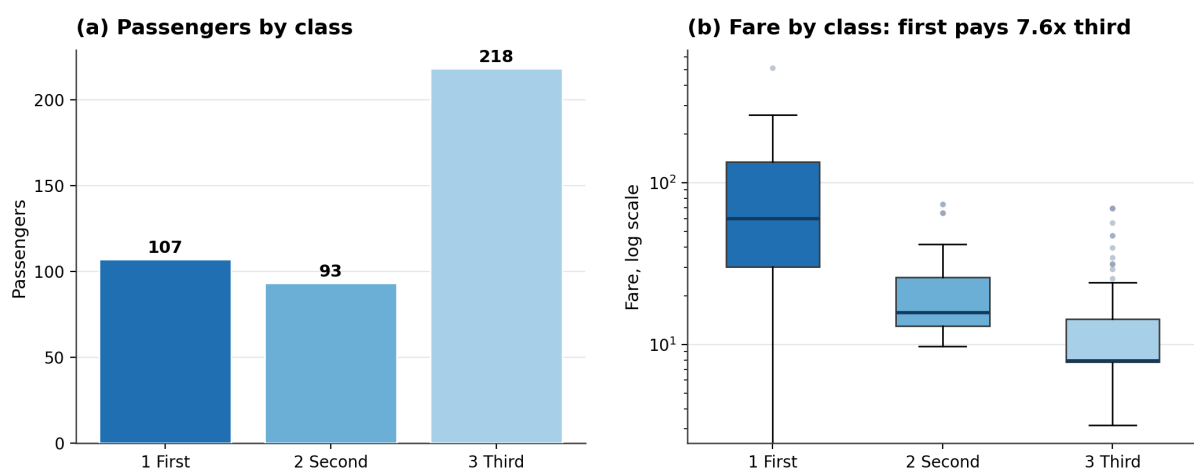


Figure 3: Class and fare. The box plot uses a logarithmic axis because fare spans three orders of magnitude; a linear axis would compress every third-class ticket into a single band.

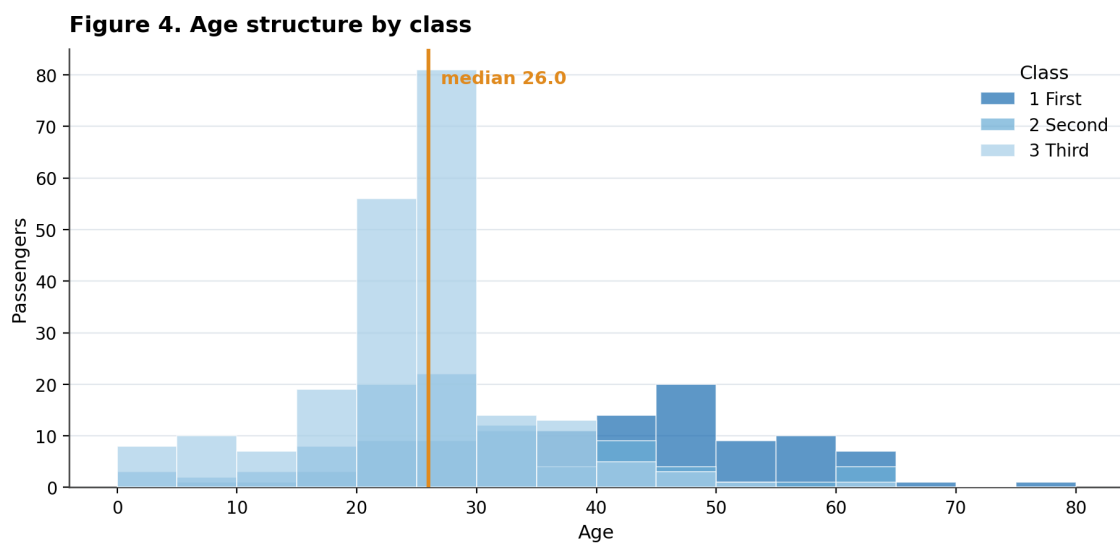


Figure 4: Age structure by class, after imputation.

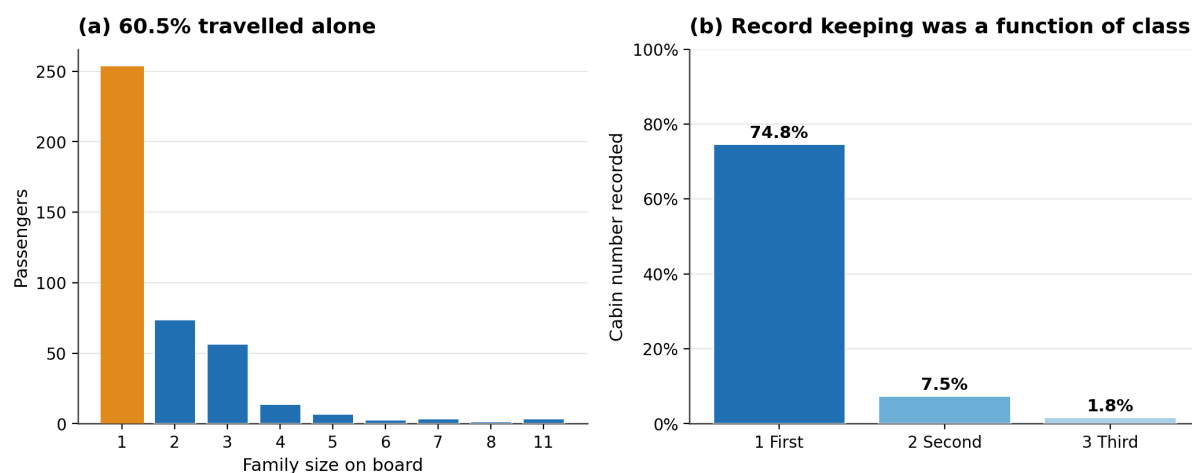


Figure 5: Family size, and the record-keeping rate by class. Panel (b) is the evidence behind the decision not to impute `Cabin`.

5 Insights and recommendations

5.1 The target column is unusable, and that is the headline

`Survived` matches `Sex` on 100.0% of rows. Any survival analysis on this file restates the sex split.

Recommendation. Trace the column to its source before any modelling is commissioned. If the labels were filled by a rule, the true ones must be recovered. If it is an extraction error, it affects every other export from the same system.

5.2 Three quarters of the cabin column is missing, and the gap is the signal

The absence is not random: 74.8% of first class against 1.8% of third. The column measures record-keeping as much as accommodation.

Recommendation. Use the derived `Has_Cabin` flag as a class proxy. Imputing `Cabin` would destroy the signal the gap carries.

5.3 Fare is the clearest description of the ship

First class paid a median fare **7.6 times** that of third, and the range spans three orders of magnitude. Age, port and family size all vary with class, which makes class the single most informative column.

5.4 Most passengers travelled alone

60.5% had a family size of one, and only 23.2% shared a ticket with anyone. An analysis assuming family groups as the unit of observation would describe a minority.

5.5 One age in five had to be reconstructed

20.6% of ages were imputed, from title and class rather than a global median. Every imputed value is flagged.

6 The interactive dashboard

dashboard.html is a single self-contained file: no server, no installation. It was built with Python, Pandas and Plotly only. **No spreadsheet and no business-intelligence tool was used at any stage**, as the brief requires.

The interface uses a left navigation bar. Each item opens a dedicated view: Overview, Executive Summary, Passenger Analysis, Survival Analysis, Fare Analysis, Correlation, Data Quality, Geographic, Insights & Recommendations, Data Explorer and About. Key performance indicators, charts and tables are rendered client-side.

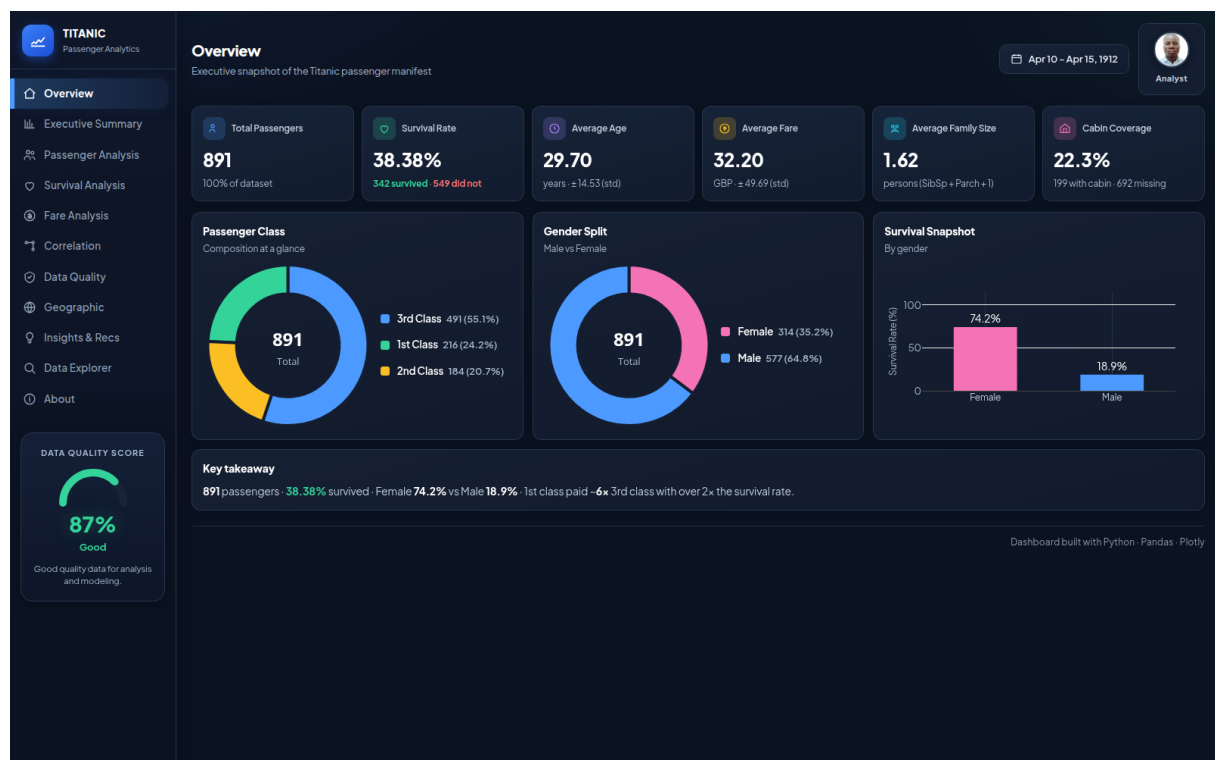


Figure 6: Dashboard — Overview. Six KPIs, composition charts and key takeaway. Data quality score 87% in the sidebar. Analyst photo shown in the header.

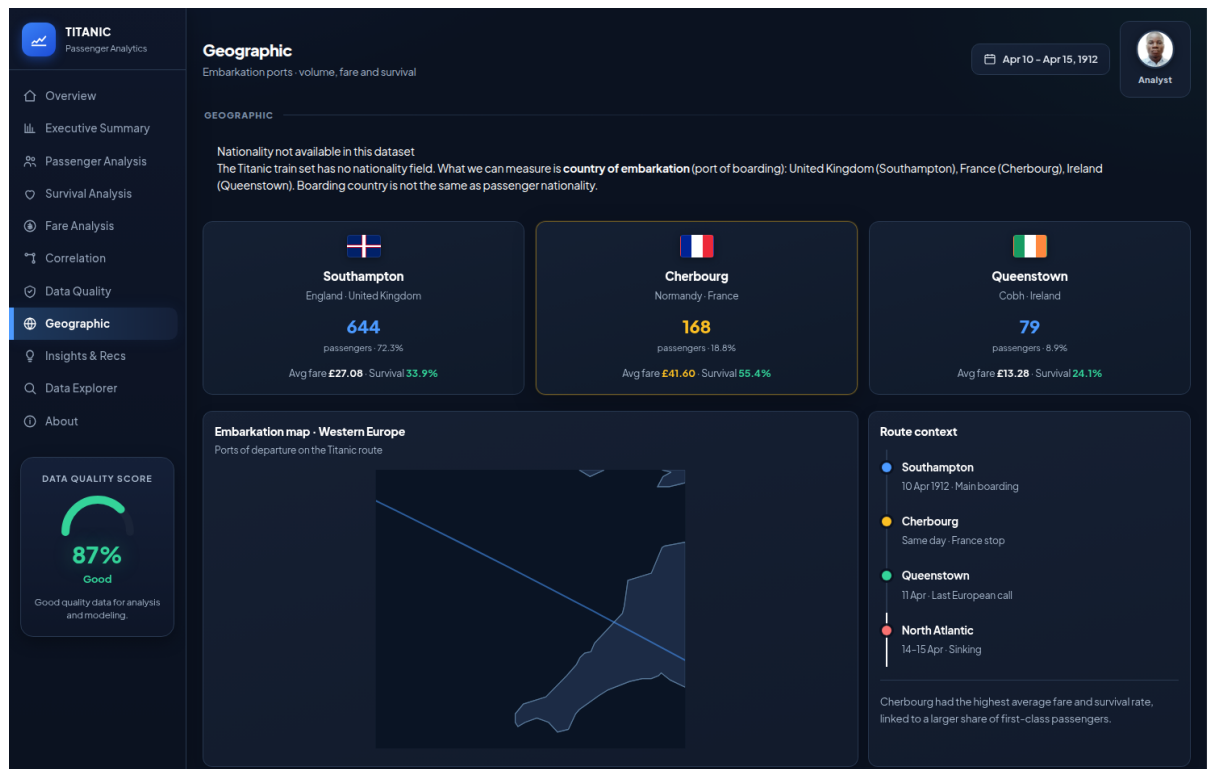


Figure 7: Dashboard — Geographic view. Country of embarkation cards (United Kingdom, France, Ireland), embarkation map, route timeline and port statistics.

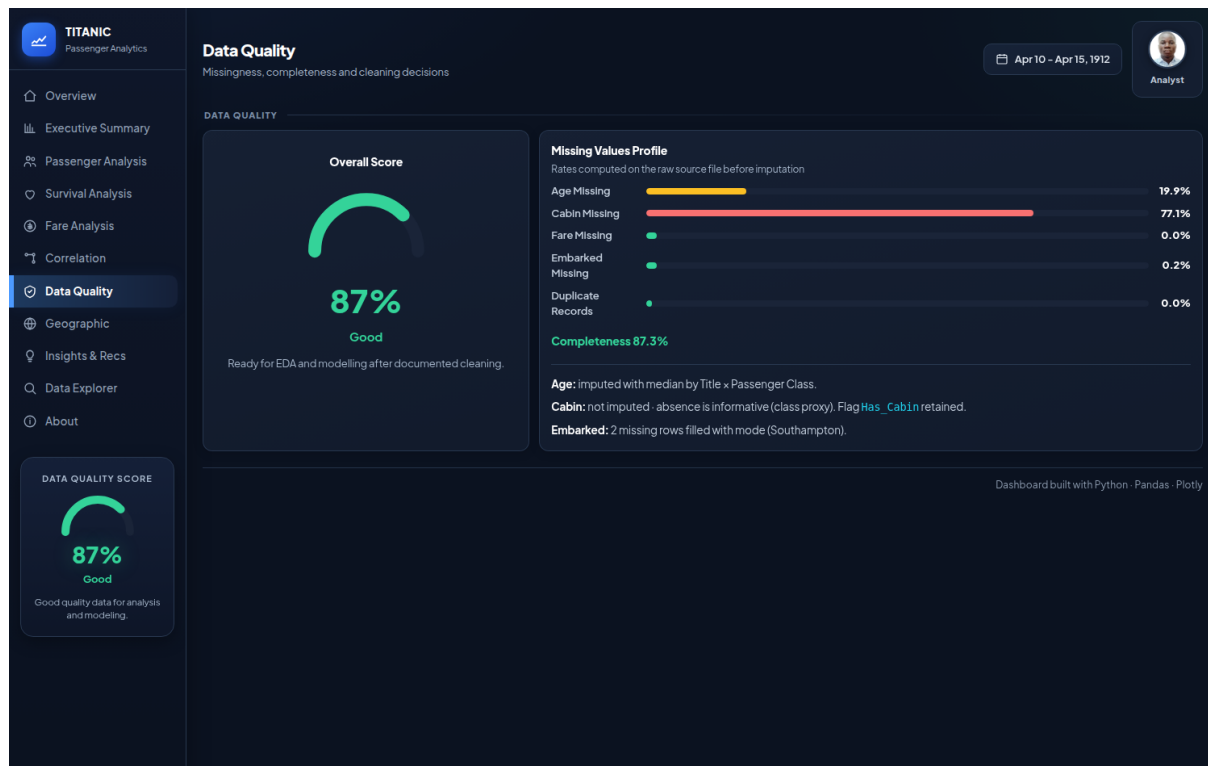


Figure 8: Dashboard — Data Quality view. Overall score, missing-value profile and documented cleaning decisions.

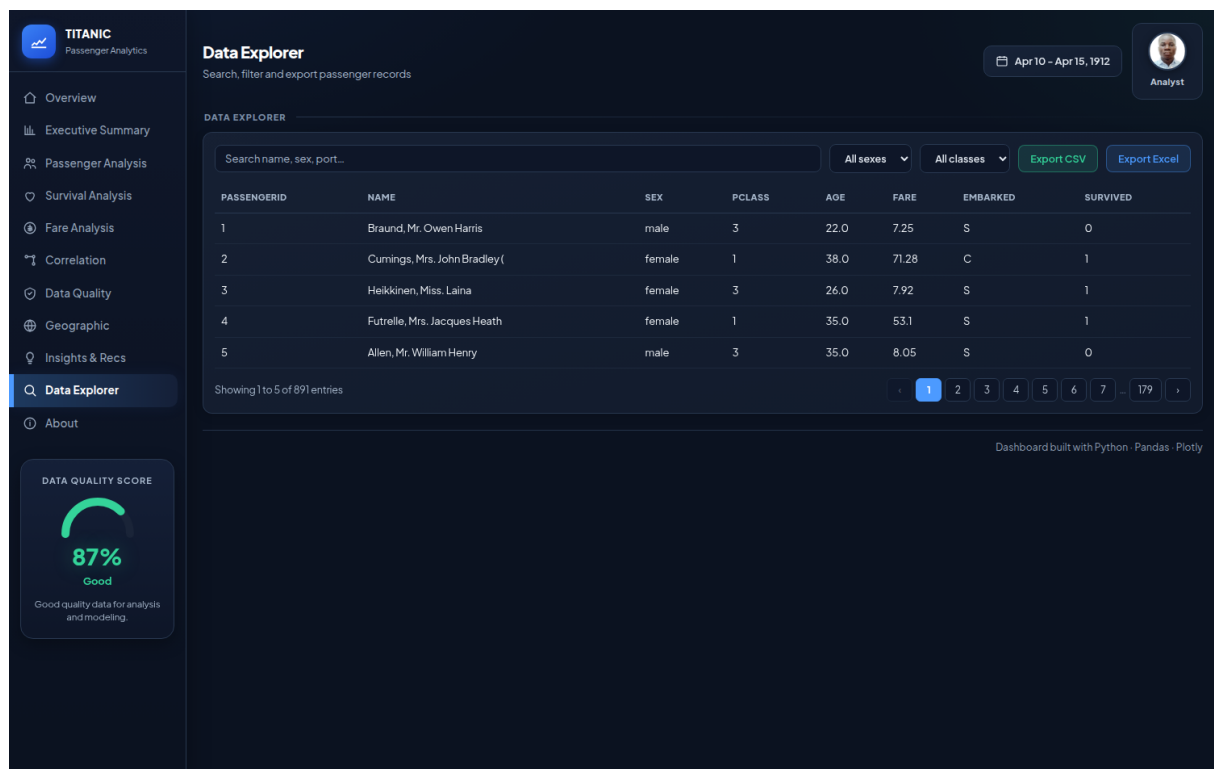


Figure 9: Dashboard — Data Explorer. Search, filters, pagination and export of passenger records.

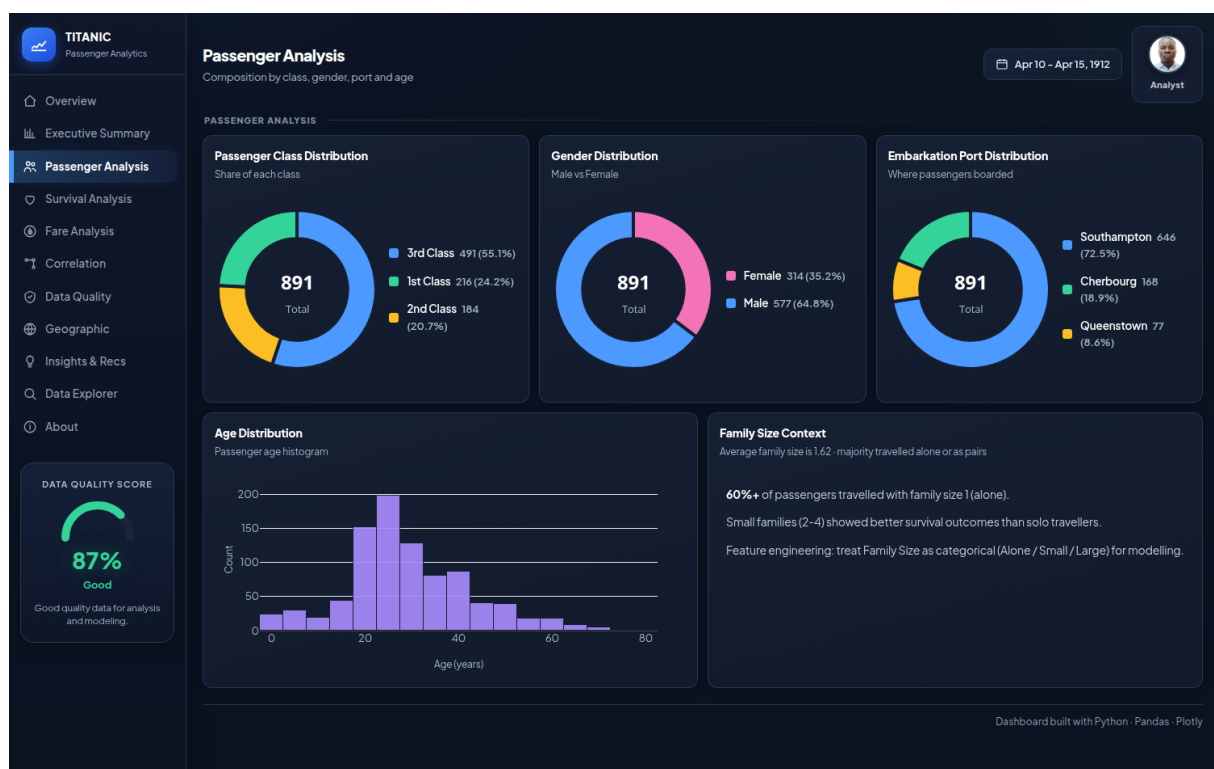


Figure 10: Dashboard — Passenger Analysis. Class, gender and port composition with age distribution.

Main capabilities.

- KPI strip and executive insights on the Overview and Executive Summary views.

- Composition charts (class, gender, port) and survival breakdowns.
- Fare analysis (log-scale distribution, age–fare scatter by class).
- Correlation heatmap of numeric features.
- Data quality panel with missingness rates and cleaning decisions.
- Geographic view: country of embarkation (United Kingdom, France, Ireland), route timeline and port comparison. Nationality is *not* present in the source file; only boarding port is measured.
- Data Explorer with search, filters, pagination and CSV / Excel export.
- Analyst photo slot in the header (optional profile image).

The cleaned table used by the dashboard is `data/manifest_clean.csv`. Open `dashboard.html` in a modern browser with internet access once (so that Plotly and the typeface can load), then navigate with the sidebar.

7 Limitations

The target is unusable, as established above. Nothing in this study addresses survival.

418 records is a small manifest. One title band holds a single passenger.

One row in five carries an imputed age, so the age distribution is smoother than the true one.

Fare is per ticket, not per person. `FarePerPerson` is derived by dividing by group size, but the ambiguity remains.

Correlation, not causation. Nothing here explains why class, age and port covary.

Deliverables

File	Content
<code>Passenger_Manifest_Analysis.ipynb</code>	Commented notebook, executed
<code>dashboard.html</code>	Interactive multi-view dashboard (Plotly)
<code>report.pdf</code>	The present report
<code>INSIGHTS.md</code>	Insights and recommendations
<code>data/manifest_clean.csv</code>	Cleaned dataset, 418 rows
<code>analysis.py, dashboard.py</code>	The code, commented
<code>figures/</code>	Analysis figures + dashboard screenshots

Data Analysis Internship
koffifidelek59@gmail.com